

Transformer Embeddings

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January 2026



Executive Summary

Current Reality

- Transformer Embeddings (TE) have consistently shown **strong performance and efficiency gains** over traditional feature engineering
- However, today's TE ecosystem is **operationally fragmented**, with separate pipelines, refresh cadences, and ownership across LOBs
- The current TE foundation was trained under **legacy constraints** and does not support scalable, governed, cross-LOB adoption

What We've Done

- Established a **unified "latest" TE access layer** across Commercial, Medicaid, and Medicare to unblock evaluation
- Created a **stable consolidated dataset for downstream users** while underlying pipelines remain heterogenous
- Laid the groundwork for **retraining on GCP**, enabling evaluation, migration planning, and future standardization

What's Next

- **Retrain TE model** on a governed, cross-LOB GCP foundation
- Align on **inference strategy, refresh cadence, and migration approach** for downstream models
- Partner with downstream teams to **validate performance, manage adoption, and sunset legacy embeddings**

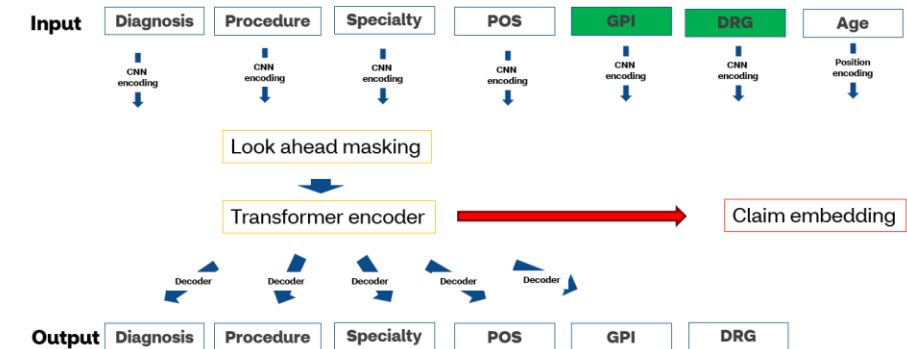
What Are Transformer Embeddings (and why we use them)

Introduction

- Transformer Embeddings (TE) **summarize a member's entire medical history** into a single, dense representation that models sequence, timing, and context – not just counts
- Medical events are treated as a **time-ordered sequence**, not independent rows
- The model learns **relationships across diagnoses, procedures, settings, and time**
- More recent and more relevant events naturally matter more

Why this is better than traditional features

- Captures **order and timing**, not just freq
- Handles **irregular, sparse medical data** naturally
- Reduces hundreds of hand-crafted features into a **single reusable signal**
- Improves performance across **many downstream models**, not just one

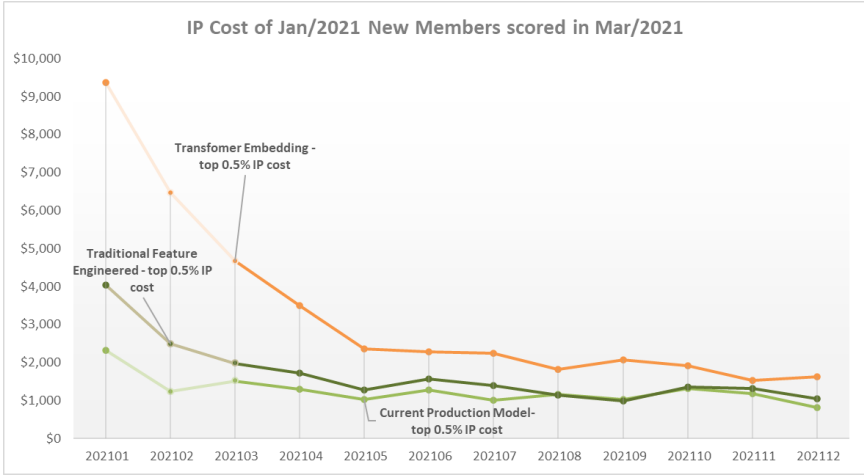


Current Evidence Snapshot

What we've seen so far across prior runs:

- TE models consistently **outperform traditional feature engineering** for high-risk member identification
- TE model identifies members **with higher IP likelihood** and **higher IP/ER** and **overall medical cost**
- Performance gains are **largest at higher-risk deciles**, where business impact is greatest
- **Standardizes feature creation from claims history**, enabling faster iteration and reducing time-to-model for new use cases

Lift at Top 1%	Commercial	Medicaid	Medicare
Traditional FE	13.6	10.2	6.1
TE only	16.4	16.9	7.3



Results from Commercial IP model

Slide courtesy of Jane Zou, Elle Palmer, Zenon Cheng

Current State: Embeddings are Fragmented across LOBs

Commercial

- Nested / array-based embedding storage
- Daily incremental + weekly full refresh
- Stored as a standalone embedding table
- Coupled with RAP model pipeline

Medicaid

- Flat, column-based embedding schema
- Monthly full refresh
- Standalone embedding schema
- Pipeline stable and refreshing as expected

Medicare

- Flat, column-based embedding schema
- Refreshed weekly with IP model scoring
- Embedding subset stored, coupled with other IP model features
- Pipeline execution constrained by compute availability and design (last successful refresh: November 2025)

Key Takeaway: Embeddings exist as LOB-specific artifacts with inconsistent schemas, refresh cadences, and operational reliability – making downstream evaluation and migration difficult

What We Did: Unified “Latest” Embedding Access Layer

Unified Access Layer

- Single consolidated Big Query view in CSDI dataset
- Brings together embeddings from Commercial, Medicaid, and Medicare
- Acts as a single access point for downstream users
- Pipelines remain owned and operated independently by LOB/model teams
- Access layer provides a stable dataset across heterogenous ownership

Design Choice

- Stores only the latest embedding per member
- Fully retrained TE product will be published to the same dataset
- Minimizes downstream change when transitioning to the unified product

Why “Latest” Only (For Now)

- Serves as a bridge, not the final TE product
- Not intended for model training or long-term storage – inference and evaluation only
- Reduces operational fragility while pipelines are heterogeneous
- Avoids unnecessary storage and compute costs during transition
- Enables fast evaluation on current state

Key Takeaway: This layer decouples downstream evaluation from pipeline ownership, simplifying access without changing underlying models, unblocking evaluation and migration planning

Access view here: `edp-prod-storage.edp_ent_transformer_srcv.embedding_combined`

Why Retraining Is Necessary: Rebuilding the Embedding Foundation

Core Reasons for Retraining

Retraining is not a routine model refresh – it is required to rebuild transformer embeddings on a governed, scalable GCP foundation that supports long-term enterprise use

- **Establish a governed and reusable foundation**
 - Current embeddings were trained on **legacy, undocumented pipelines**
 - Training data lineage was not preserved during migration to GCP
- **Improve coverage and population representativeness**
 - Inclusion of **Medicaid** for full population representation
 - Better generalization across **diverse member cohorts**
 - Alignment with **current utilization patterns**, not legacy behavior

Dimension	Current State (V2)	Retrained State (V3)
Data Coverage	Commercial, Medicare from 2016	All LOBs – Commercial, Medicare, Medicaid from 2023
Infrastructure	Legacy Hadoop, undocumented	Fully migrated and governed on GCP
Coverage	Limited data sources	Expanded to include GPI, DRG and National Provider Taxonomy
Governance	None	Full lineage, reproducibility, and metadata tracking

Key Takeaway: To move from a shared access layer to a true shared product, retraining is required to ensure governance, representativeness, and long-term scalability

Training Enablement, GPU Readiness & Cost Guardrails

Why this matters

- Transformer retraining at enterprise scale requires H100-class GPUs
- Not feasible on Vertex AI default GPUs (T4) given model size + runtime constraints
- T4 incompatibility with standard and mainstream model architectures and algorithms

What had to be aligned

- **Business Justification** (why H100 vs alternatives)
- **Training scope & time bounds** (not open-ended compute)
- **Quota + region availability** on governed GCP projects
- **Cost visibility and approvals** before execution

Machine Type	Configuration	Cost	Est. Runtime	Est. Cost	Notes
a3-highgpu-4g	4 X H100	\$44/hr	~24-36 hrs	\$1k-\$1.5k	Bounded run, preferred
n1-standard-32	4 x T4	\$3/hr	~900-1200 hrs	\$2.7k - \$3.6k	Memory/time constraints

[Link to Google Cloud Pricing Calculator](#)

Target Inference & Serving Pipeline

Inference Generation

- Retrained embedding model is used for **batch inference**
- Runs on a **defined cadence or event trigger** (claims-based)
- Embeddings are generated **as of a specific point in time**

Embedding Storage

- Inference outputs are written to **CSDI** dataset
- Stored as **time-anchored snapshots** (as of index date)
- Enables:
 - Evaluation over a fixed cohort
 - Reproducibility of results
 - Auditability

Downstream Consumption

- Downstream models access embeddings via a **stable CSDI interface**
- Consumers are decoupled from pipeline ownership

Operational Guardrails

- Inference jobs are **auditable and executable**
- Backfills and re-runs are **supported without retraining**

Key Takeaway: This pipeline supports a production TE product with phased rollout, ensuring stability and controlled adoption

Embedding Refresh Cadence – Operating Model Decision

Option 1 : Fixed Time-Based Refresh

Weekly or monthly refresh for all members

- Pros
 - Easy to schedule and monitor
- Tradeoffs
 - Recomputes embeddings even when nothing has changed
 - Higher ongoing compute cost

Option 2: Event-Driven Refresh (Recommended Direction)

- Weekly refresh for members with new claims
- Monthly full refresh to capture time-based effects (e.g., aging, recency decay)
- Pros
 - Aligns embeddings with meaningful member state changes
 - Captures temporal dynamics even without new claims
 - Avoids unnecessary recomputation for static members
 - Scales better as population grows
- Tradeoffs
 - More complex orchestration

This is a product operating design that impacts cost, scalability, and downstream stability – and should be aligned across teams

Migration & Adoption Strategy – Open Questions for Alignment

Current Reality

- 3 embedding pipelines running today
 - Different owners, refresh cadences
- New embedding version implies downstream model retraining
- We cannot switch embeddings without coordination

Proposed Initial Adoption Scope

- Start with a small set of downstream models to validate
 - Commercial IP Model
 - Medicaid IP Model
 - Medicare IP model (test Komodo value-add)

Key Questions for Alignment

- **Evaluation/Validation Ownership**
 - A: TE Product Team
 - B: Individual model owners
- **Once adoption criteria are met, how should legacy embeddings be handled?**
 - A: Define a fixed overlap window, then deprecate
 - B: Deprecate per LOB as they migrate
 - C: Keep legacy embeddings indefinitely (not recommended)



Model Architecture & Training Ablations

Model Architecture and Refactoring

Preserve hierarchical structure

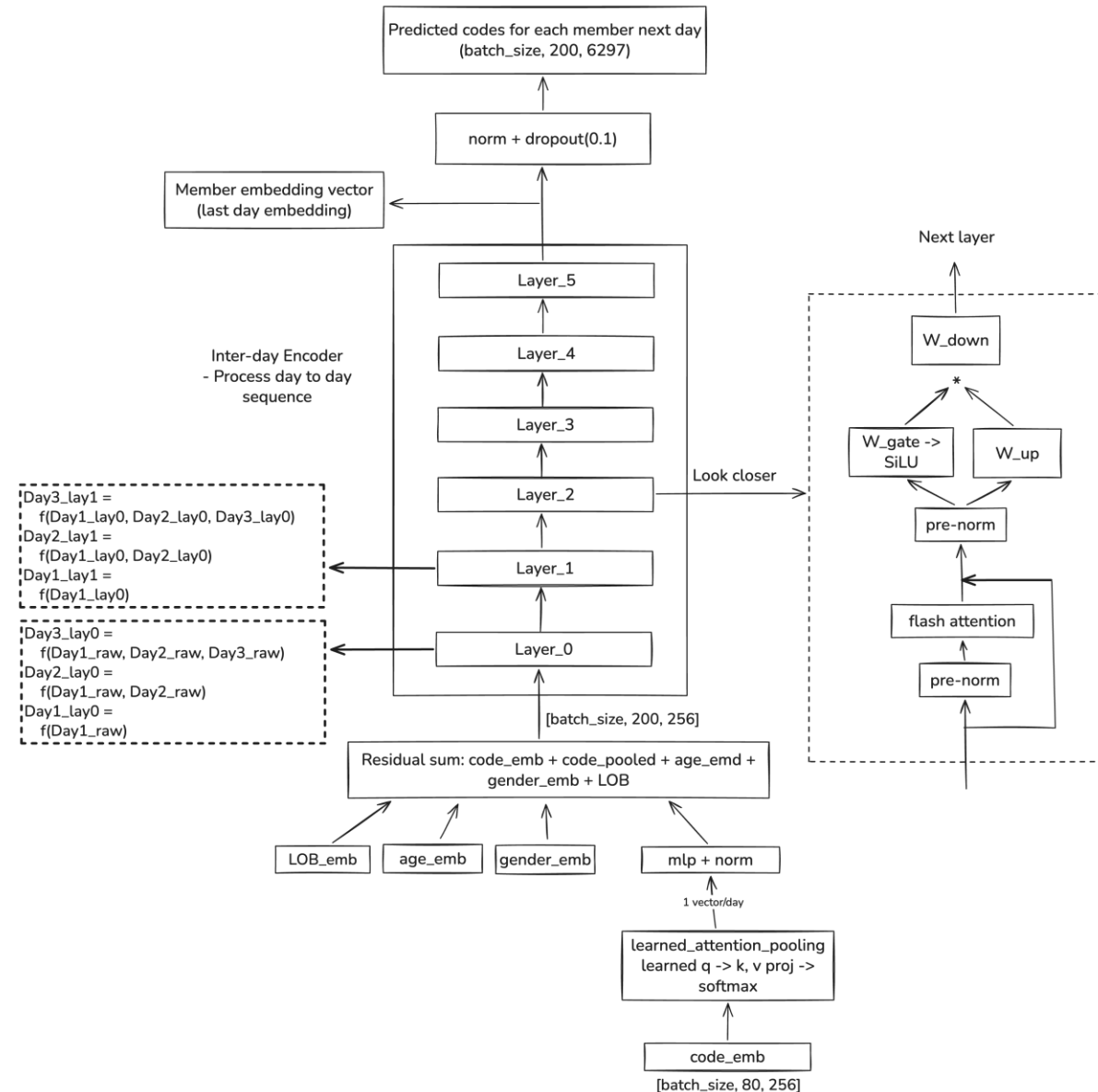
- **Hierarchical structure** for encoding within- (80 codes) and inter-day (200 days)
- Trained to predict **next-day medical codes** (multi-label, 6,297 target groups)

Training & inference efficiency

- Replace the attention layer with **flash attention (FA)** to avoid GPU IO bottleneck and provides faster training with less GPU memory
- Replace the regular FFN with SwiGLU FFN for stable training and context-aware feature selection

Optimize training strategy

- Weighted BCE loss accounting for rare codes
- Enable mixed precision training
- SGD -> Adam learning rate optimizer
- Head dimension 16 -> 32 (fit flash attention)



Preliminary Ablation Results

Data

- Commercial: 1.06M
- Medicare: 0.56M
- Medicaid: 0.13M
- Day count > 10
- Time period: 1/1/2023 – 12/31/2023
- Claim lookback period: 36 months with 90d delay

- Full input codes: 75.51k
- Target codes: 6.30k

Ablation config.:

- Legacy baseline: Dense + legacy config
- Exp2: Dense + optimized config (OC)
- Exp3: Dense + OC + flash attention (FA)
- Exp4: MOE + FA + OC
- One epoch

Infrastructure:

- GPU: T4 * 4 (16GB HBM each)
- Machine type: n1-highmem-64
- Data parallelism applied

Observations

- Optimized training strategy achieves over **2x** improvement in prediction performance
- Flash attention reduced training cost by **34%** and reduced GPU memory usage by **61%**
- MOE matched dense model performance
- **Notice:** intrinsic metrics captures only internal metrics for training quality and outcomes on predicting next day's labels; **downstream evaluation** provides definite and business-driven measure of the embedding quality.

Ablation config.	Recall @10 ¹	Precision @10 ²	Micro Recall@10 ³	NDCG @10 ⁴	Training time (hrs)	GPU peak memory (GB)
Random reference	0.007	0.001	0.002	0.001	NA	NA
Legacy model (baseline)	0.579	0.120	0.234	0.169	6.15	28.29
Optimized model	0.825	0.235	0.478	0.401	9.65 ⁵	20.58
Optimized + FA	0.828	0.237	0.462	0.398	4.09	11.13
Optimized + FA + MOE	0.827	0.237	0.461	0.399	4.74	13.39

1. Recall@10: fraction of members have at least 1 correct code in top-10 predictions
2. Precision@10: fraction of correct predictions at top-10 predictions
3. Micro-recall@10: fraction of true codes found in top-10 predictions
4. NDCG@10 (normalized discounted cumulative gain at top-10), indicates ranking quality, where 1 is perfect and 0 is none is relevant at top10 predictions
5. Due to the constraints of GPU memory and use of data parallelism, this experiment used 16 batch size instead of 32. The training time, given more advanced GPU, should be less than 6 hours

INTERNAL USE AND NOT FOR DISTRIBUTION OR RECIRCULATION

Lesson learnt, downstream evaluation planned

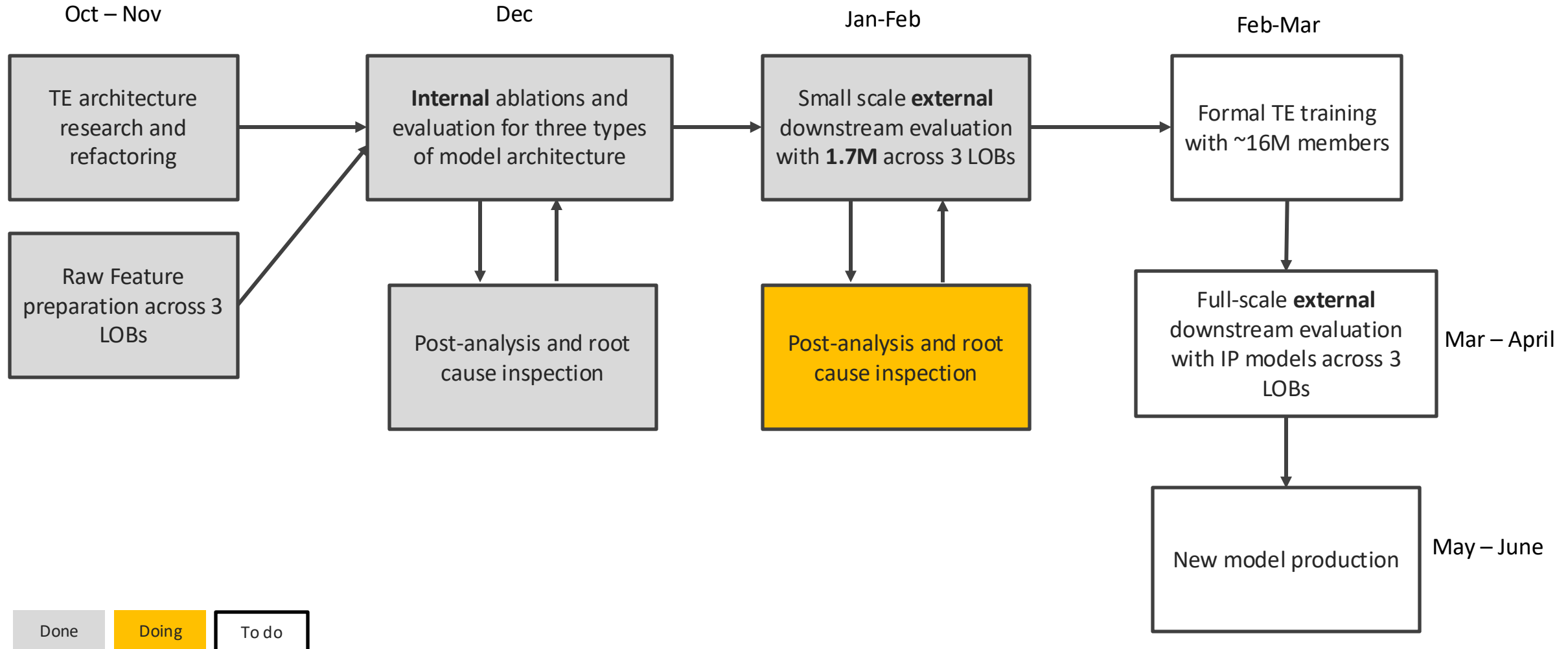
Lesson learnt

- Flash attention significantly improves the training efficiency, promising for longer patient histories with minimal computational overhead
- MOE provided marginal benefits to the model performance at current model scale; downstream task will inform final decision
- Computational resources remain a key consideration for LLM (even small) pretraining and post-training; cost estimation, governance approval and resource reservation are a must

Next step: downstream evaluation across LOBs

- Population: out of bag members (not in transformer pretraining data)
- Sampling strategy: sampled members to generate embeddings for quick turnaround (e.g., 30%)
- Tasks across three LOBs: Commercial IP, Medicare IP and Medicaid IP
- Downstream modeling and pipeline: following production model development procedures

Milestones and timeline



Downstream evaluation (small scale) – Commercial and Medicare IP

Major findings:

1. The TE with optimizations in general outperforms the legacy TE , validating architectural and training improvements
2. The IP model trained with TE embedding features achieve performance comparable to traditional features, demonstrating that TE captures most of the predictive signal used in production
3. The comparative advantage of TE embeddings varied across LOBs

	Production IP model	Legacy-TE		Optimized-TE		Optimized-FA-TE		Optimized-FA-MOE -TE	
Feature type	Engineered feature	Hybrid	Embedding	Hybrid	Embedding	Hybrid	Embedding	Hybrid	Embedding
Medicare IP model									
AUC ROC	0.742	0.747	0.685	0.761	0.746	0.758	0.736	0.759	0.746
AUC PR	0.199	0.204	0.15	0.217	0.201	0.21	0.186	0.21	0.197
Lift@1%	6.201	6.323	4.864	6.769	6.35	6.391	5.742	6.161	5.985
Commercial IP model									
AUC ROC	0.832	0.817	0.724	0.821	0.783	0.824	0.79	0.823	0.786
AUC PR	0.1	0.099	0.035	0.089	0.067	0.105	0.078	0.107	0.076
Lift@1%	18.682	18.566	8.561	17.677	14.608	19.082	15.059	18.463	16.091

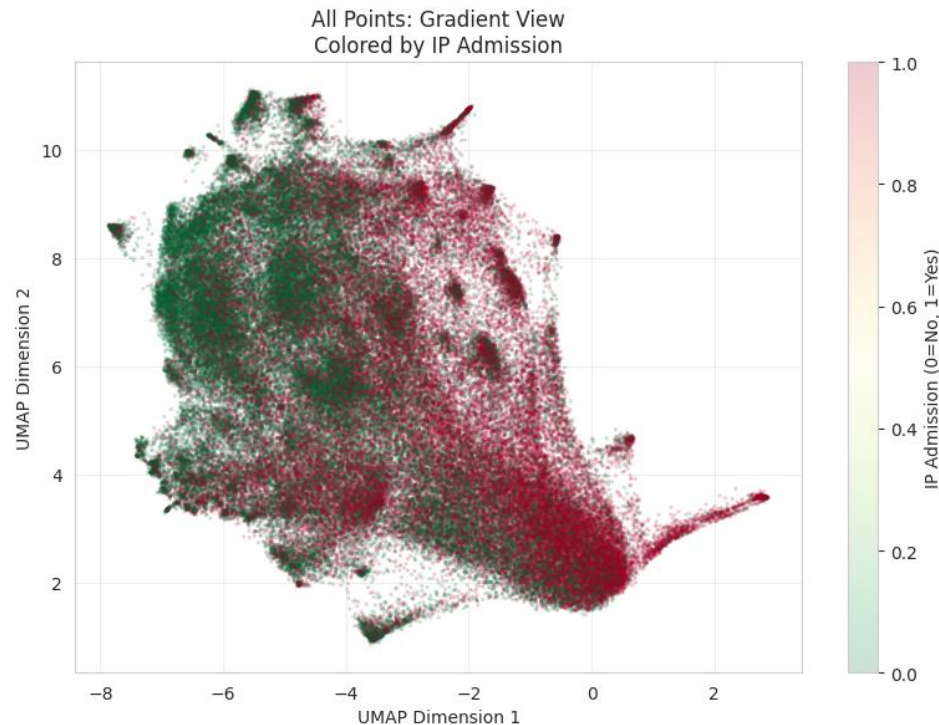
Experiment setups:

- TE training data size: 1.75M in 2023
- 4 T4 GPUs; 1 Epoch.

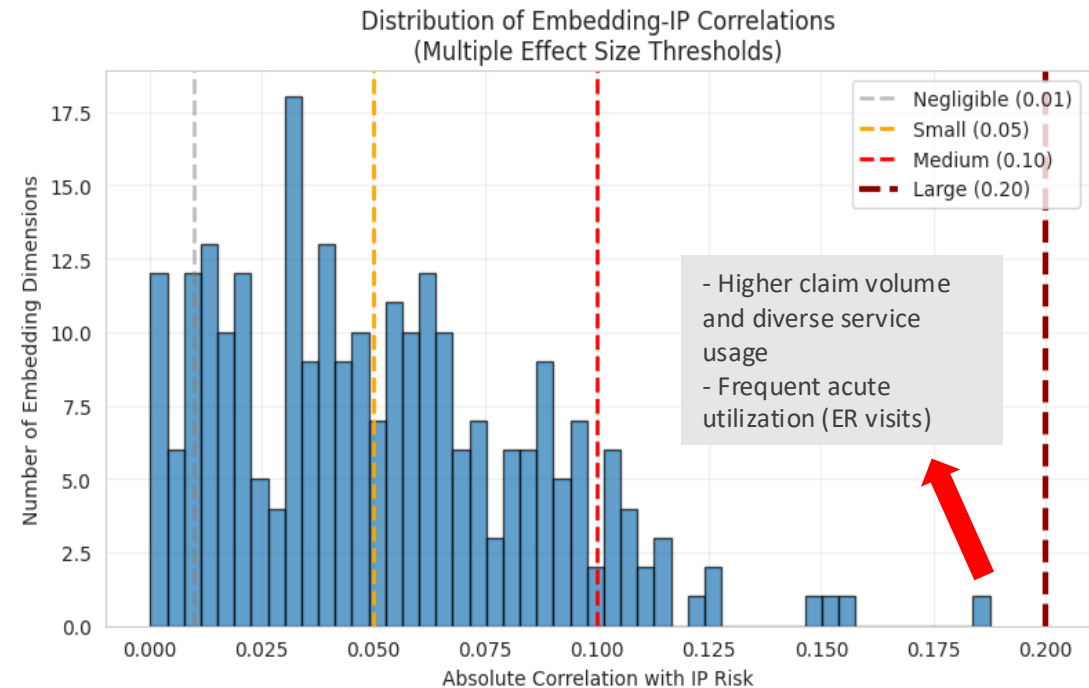
Deep dive 1 – Does the embeddings capture the right information?

Major findings:

1. **Embeddings encode clear separation between high-risk and low-risk IP members**, indicating clinically relevant structure
2. Out of 256 dimensions, over 37% of dimensions are moderately relevant to the IP risk and 9% are highly relevant, confirming embeddings are not noisy or redundant representations



- Centroid distance: 1.506
- Cohen's D effect size: 0.691 (considered medium effect, noticeable separation)



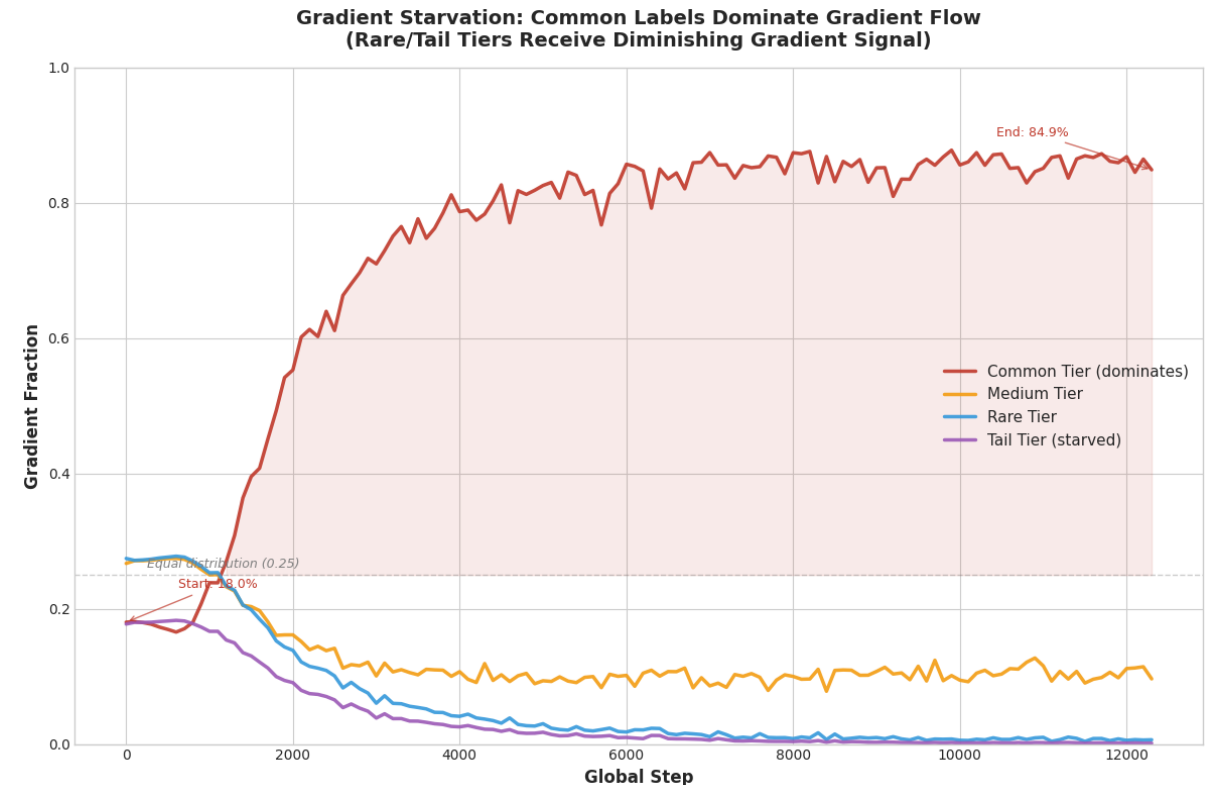
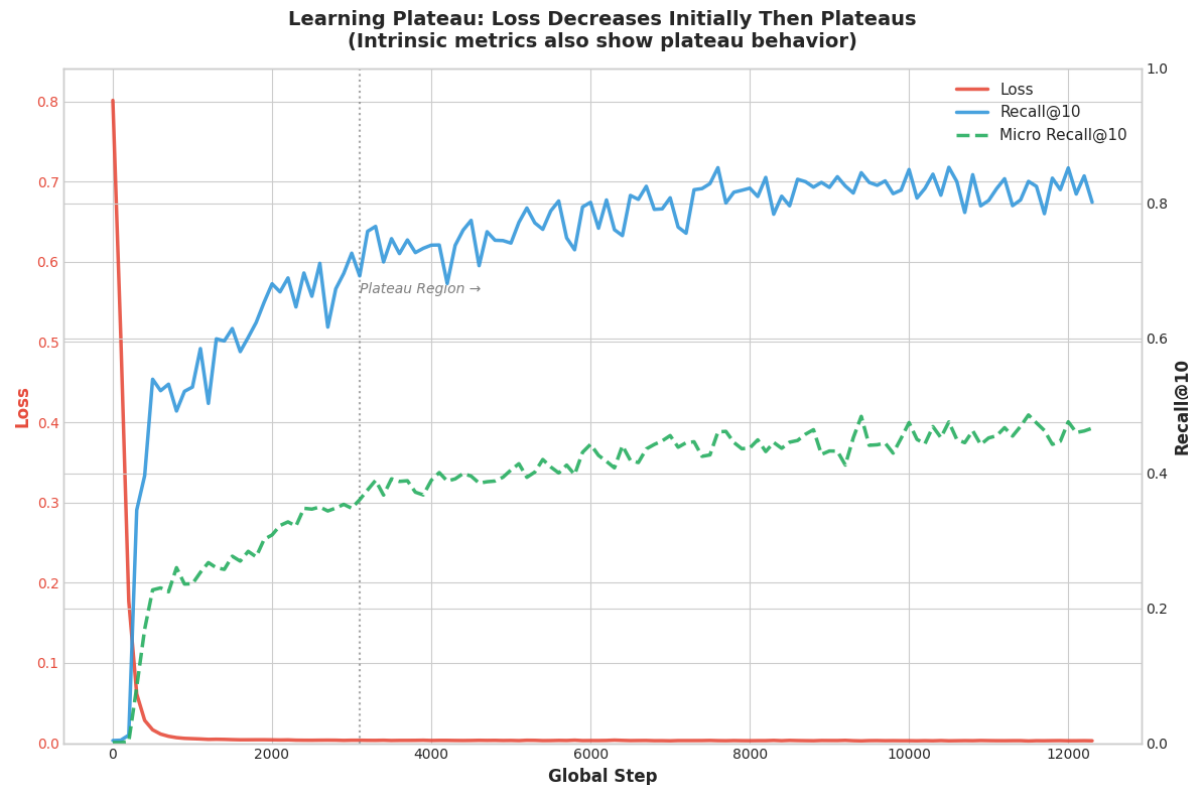
- $N(|r| \geq 0.05) = 119$ (46%)
- $N(|r| \geq 0.01) = 23$ (9%)

Deep dive 2 – Did TE really learn what it should learn?

Major finding:

1. TE model learned very fast at the beginning and then plateaus
2. Common codes are “hogging” all the learning and starving the rest of the codes.

This behavior is expected in highly imbalanced medical vocabularies and known training dynamics, highlighting optimization opportunities rather than limitations of the embedding approach



Discussion: Interpreting Results & Path Forward

What we learned

- Optimized transformer embeddings **capture clinically meaningful signal** and generalize to downstream prediction tasks
- **At current scale, embeddings achieve parity with production-engineered features**, which are heavily task and domain specific
- Obtained a **better explainability** of the learning mechanism of the TE and clinical meaning of the embeddings

Why embeddings did not outperform tabular features

- **Hypothesis 1: Training data scale**
 - Current evaluation uses ~1.75M members (~10% of total population across 3 LOBs)
 - Representation learning performance is known to improve non-linearly with data scale
- **Hypothesis 2 (Confirmed): Embeddings are intentionally task-agnostic**
 - They are trained to learn reusable clinical representations, rather than maximize performance on a single downstream task
- **Hypothesis 3 (Confirmed): Uncommon codes are under-represented**
 - Severe class imbalance concentrates learning on common codes

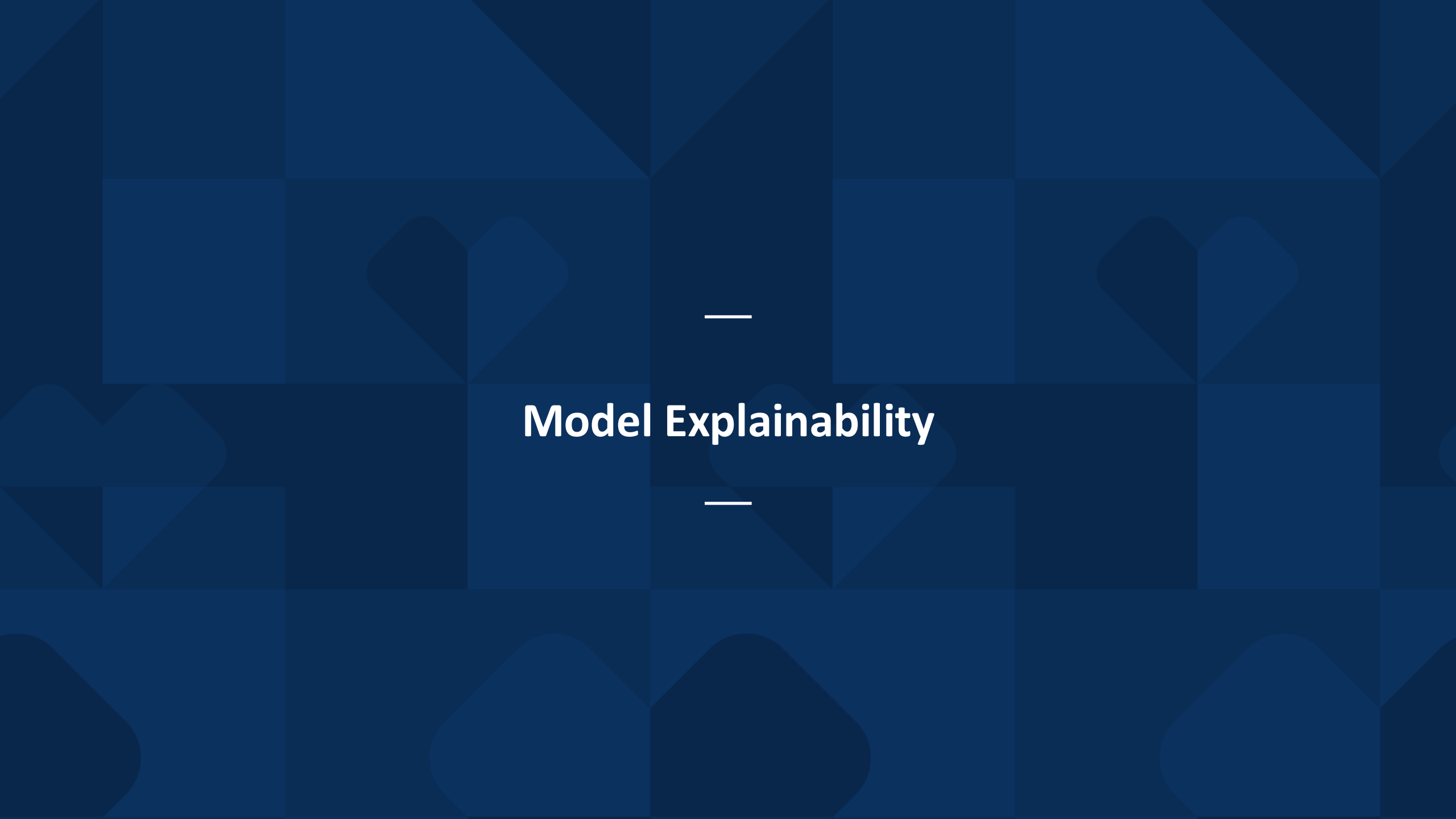
Path Forward

- **Apply targeted optimization** to improve rare-code learning
- **Embedding dimension explainability**
- **Scale training data across LOBs** to improve representation depth

Downstream evaluation

Test	Production IP model	Production TE		New TE	
Feature type	Engineered feature	Hybrid	Embedding-only	Hybrid	Embedding-Only
Medicare IP model					
AUC ROC		0.758	NA	0.765	0.767
AUC PR		0.233	NA	0.236	0.226
Lift@1%		6.83	NA	7.13	6.33
Lift@10%		3.53	NA	3.6	3.51
Commercial IP model					
AUC ROC	0.85	0.849	0.799	0.858	0.828
AUC PR	0.122	0.124	0.088	0.112	0.08
Lift@1%	22.506	22.057	18.911	23.041	18.804
Lift@10%	5.998	5.983	5.114	6.123	5.489
Medicaid IP model					
AUC ROC					
AUC PR					
Lift@1%					
Lift@10%					

Test	Production IP model	Production TE		New TE	
Feature type	Engineered feature	Hybrid	Embedding-only	Hybrid	Embedding-Only
Medicare IP model					
AUC ROC		0.758	NA	0.764	0.761
AUC PR		0.217	NA	0.217	0.206
Lift@1%		6.99	NA	6.92	6.36
Lift@10%		3.57	NA	3.58	3.5
Commercial IP model					
AUC ROC	0.85	0.846	0.8	0.85	0.824
AUC PR	0.122	0.109	0.08	0.106	0.081
Lift@1%	22.506	22.45	19.052	22.351	19.095
Lift@10%	5.998	5.969	5.208	5.991	5.444
Medicaid IP model					
AUC ROC					
AUC PR					
Lift@1%					
Lift@10%					



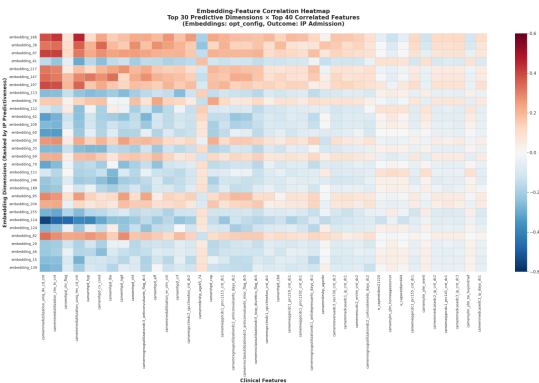
Model Explainability

Review of previous work in TE explainability

Task-oriented

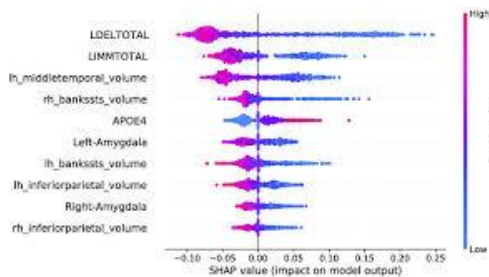
Semantic meanings of the embedding

Correlation between a given TE dimension with traditional features, e.g., emb154 is associated with diabetes in Medicaid; emb186 encodes claims volume and utilization patterns



Primary driver of predictions

Identify **importance of specific embedding dimensions** on downstream classification tasks by measuring effects of removing such dimensions



Primary driver of prediction – LLM

Use **LLM** to directly generate reasoning and explanations of the prediction outcomes, given the raw claim inputs and derived feature importance.

Member A:

Service Date: 2022-12-30
icd9_dx_cd: I10,G93.40
prodr_cd: 84443,82607,82746,84439
hcfa_plc_srv_cd: 81
taxonomy codes: 291U00000X
...
Service Date: 2023-01-05
icd9_dx_cd: I10,E11.621
prodr_cd: 99309
hcfa_plc_srv_cd: 31
...

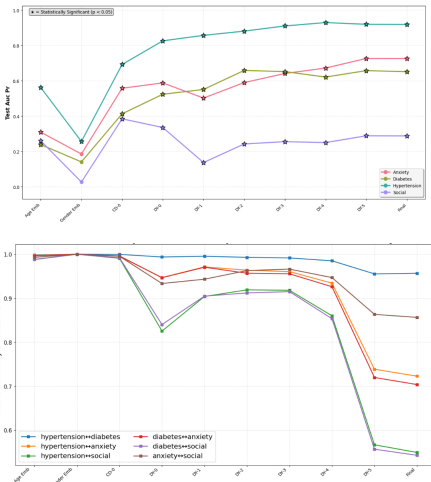


Gemini: “The member faces a high likelihood of inpatient admission in the next three months due to several compounding clinical factors.....”

Task-agnostic (Model behavior)

Model learning dynamics and concept representations

Dissect the transformer model **layer-by-layer** to understand what clinical concepts are learned, where they are encoded, and how the learning information evolves through the network's depth



Semantic meanings of embeddings with correlation

- **Embedding 224 is associated with nearly everything**
 - Most statistically significant association is major chronic count
 - Directionality check shows higher value indicative of greater disease burden
 - Strong future IP risk associations has been found in all LoBs
- **Embedding 154 is the top embedding associated with diabetes across LoBs**
 - Directionality check shows that a lower embedding value is indicative of having diabetes (Medicaid shown)

emb154_n <-1	-1<=0	0 to 1	> 1
DIA			
0	0.0%	13.9%	76.0%
1	0.2%	63.8%	35.8%

emb224_n <-1	-1<=0	0 to 1	> 1
major_chronic_cnt			
0	1.6%	76.6%	21.6%
1	0.5%	64.6%	34.6%
2	0.1%	49.1%	50.1%
3	0.0%	35.2%	63.2%
4	0.0%	23.9%	73.1%
5	0.0%	15.6%	78.8%
6	0.0%	9.8%	81.3%
7	0.0%	5.9%	80.3%
8	0.0%	3.7%	76.6%
9	0.0%	2.3%	71.1%
10	0.0%	1.3%	64.7%
11	0.0%	0.8%	57.0%
12	0.0%	0.4%	49.4%
13	0.0%	0.3%	41.0%
14	0.0%	0.2%	34.0%
15	0.0%	0.1%	28.2%
16	0.0%	0.0%	23.8%
17	0.0%	0.0%	18.9%
18	0.0%	0.1%	14.0%
19	0.0%	0.1%	12.3%
20	0.0%	0.0%	8.6%
21	0.0%	0.0%	6.9%
22	0.0%	0.0%	6.8%
23	0.0%	0.0%	4.9%
24	0.0%	0.0%	5.8%
25	0.0%	0.0%	6.4%
26	0.0%	0.0%	0.0%
27	0.0%	0.0%	0.0%
28	0.0%	0.0%	0.0%
29	0.0%	0.0%	0.0%
31	0.0%	0.0%	0.0%

Primary driver of prediction – LLM

Claim history into TE

Service Date: 2022-12-30
icd9_dx_cd: I10,G93.40
prcdr_cd: 84443,82607,82746,84439
hcfa_plc_srv_cd: 81
taxonomy codes: 291U00000X
...
Service Date: 2023-01-05
icd9_dx_cd: I10,E11.621
prcdr_cd: 99309
hcfa_plc_srv_cd: 31
...

+

Other individualized top non-TE features by SHAP

of claim lines in the past 6 months: 20.86,
age: 80.0,
of PCP visits in the past 3 months: 1.0,
of days from recent rx to index_dt: 20.0,
Dementia condition: 1.0
...



Gemini
Flash 2.5 &
prompt

Reasoning narrative:

The member faces a high likelihood of inpatient admission in the next three months due to several compounding clinical factors.

- **Complex Diabetes with Complications:** Uncontrolled Type 2 Diabetes, evidenced by recurrent foot ulcers, cellulitis, abscesses, and osteomyelitis, presents a continuous risk for severe infections and surgical needs. The recent inpatient stay for these issues underscores this vulnerability.
- **Advanced Cognitive Impairment:** Diagnoses of Alzheimer's disease, metabolic encephalopathy, and altered mental status increase the risk of self-neglect, medication non-adherence, falls, and inability to manage acute symptoms, often necessitating hospitalization.
- **Recurrent Infections and Dehydration:** Episodes of C. difficile enterocolitis and urinary tract infections, coupled with dehydration, indicate a fragile immune system and poor physiological reserve, making the member susceptible to acute medical crises.
- **Multisystemic Disease:** The combination of cardiovascular disease, cognitive decline, and chronic infections creates a complex clinical picture where an exacerbation in one system can quickly lead to decompensation requiring acute hospital care.

Model behavior with probing and CAV – what clinical concepts are learned and where

- Key clinical concepts are captured well over the architecture
 - The concept information accumulates over layers and the capability varied across the clinical concepts

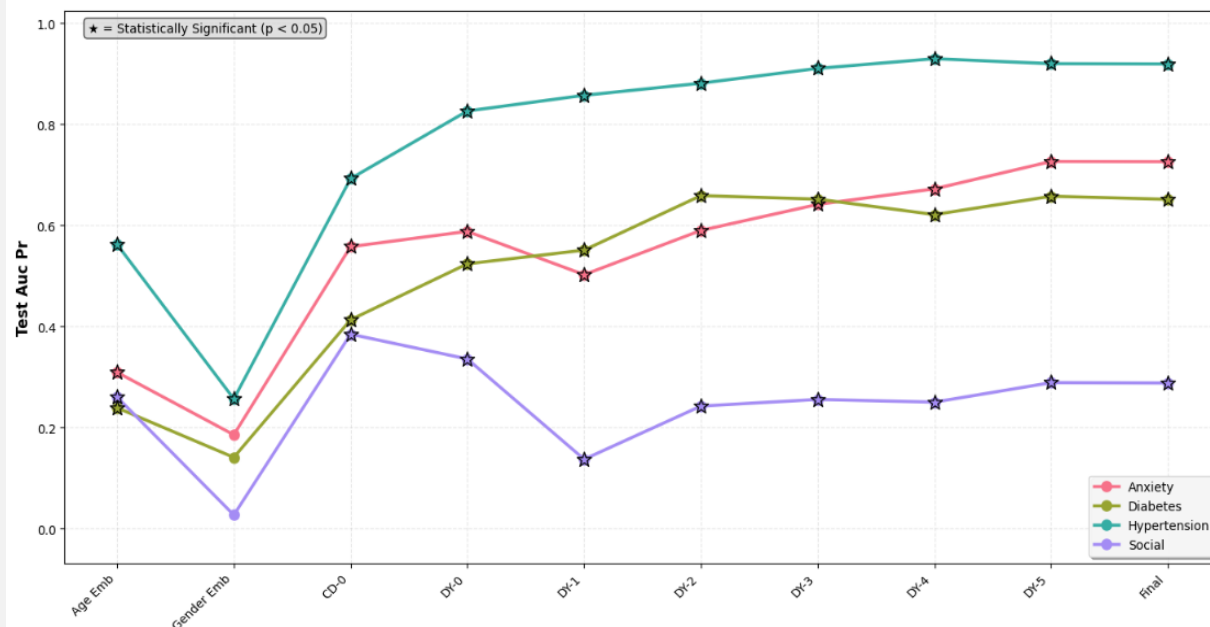


Figure 1: Performance of the transformer encoding different clinical concepts, indicated by probe classifier AUC-PR in predicting the presence of a concept within a Medicaid sampled population.

- The transformer encodes both shared and specialized information about concepts
 - It starts to differentiate different clinical concepts approaching the end of layers

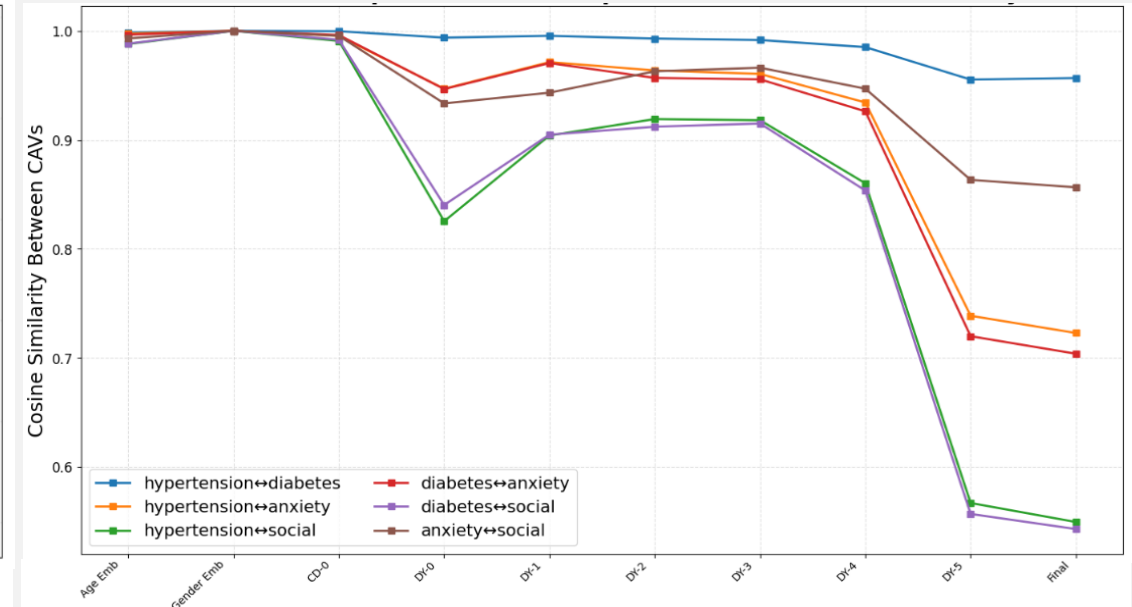


Figure 2: Cosine similarity between clinical concept vectors, indicating how much specialized information each layer encodes for a concept.

TE explainability framework

What stakeholders really want to understand?

- Semantic meanings of the embedding dimensions and the impact of each dimension on the downstream classification tasks
- TE learning progression and how it builds the clinical understanding internally
- The "explainability" means differently to DS to CM; extra efforts are needed when communicating with CMs.

What principles we should follow?

- **Faithfulness:** The explanation should accurately reflect the models' true behavior, not post-hoc rationalization; and align with domain knowledge and expectations
- **Consistency:** Similar inputs yield similar explanations across contexts (i.e., cross population, upstream-downstream)
- **Transparency:** The transparency should be operated at three levels, i.e., algorithm, procedure and limitations; caution on LLM-based explanation
- **Completeness:** Explanations should capture a sufficient portion of the outputs and model behaviors

What methods we need?

- Simple correlation with traditional features with directional check
- Task-driven methods include Shapley value, LIME
- Model behaviors methods include test with concept activation vector (TCAV), linear probing, attention mechanism visualization
- Experts' generated meaning clinical concepts and ground truth

Method 1 – Concept Activation Vectors (CAV)

Objective

- Make the embeddings interpretable using Concept Activation Vectors (CAV) to identify which embedding dimensions encode clinically meaningful information

Method

- Defined **74 claims-based clinical concepts** across 9 categories – *Chronic Conditions, Cardiovascular Conditions, Neurological & Mental Health, Cancer & Complex Care, Acute Events, Utilization Patterns, Risk Factors, Care Coordination and others*
- Generated member-level labels from the same data used to train the transformer
- Used CAVs to test which embedding dimensions predict each concept

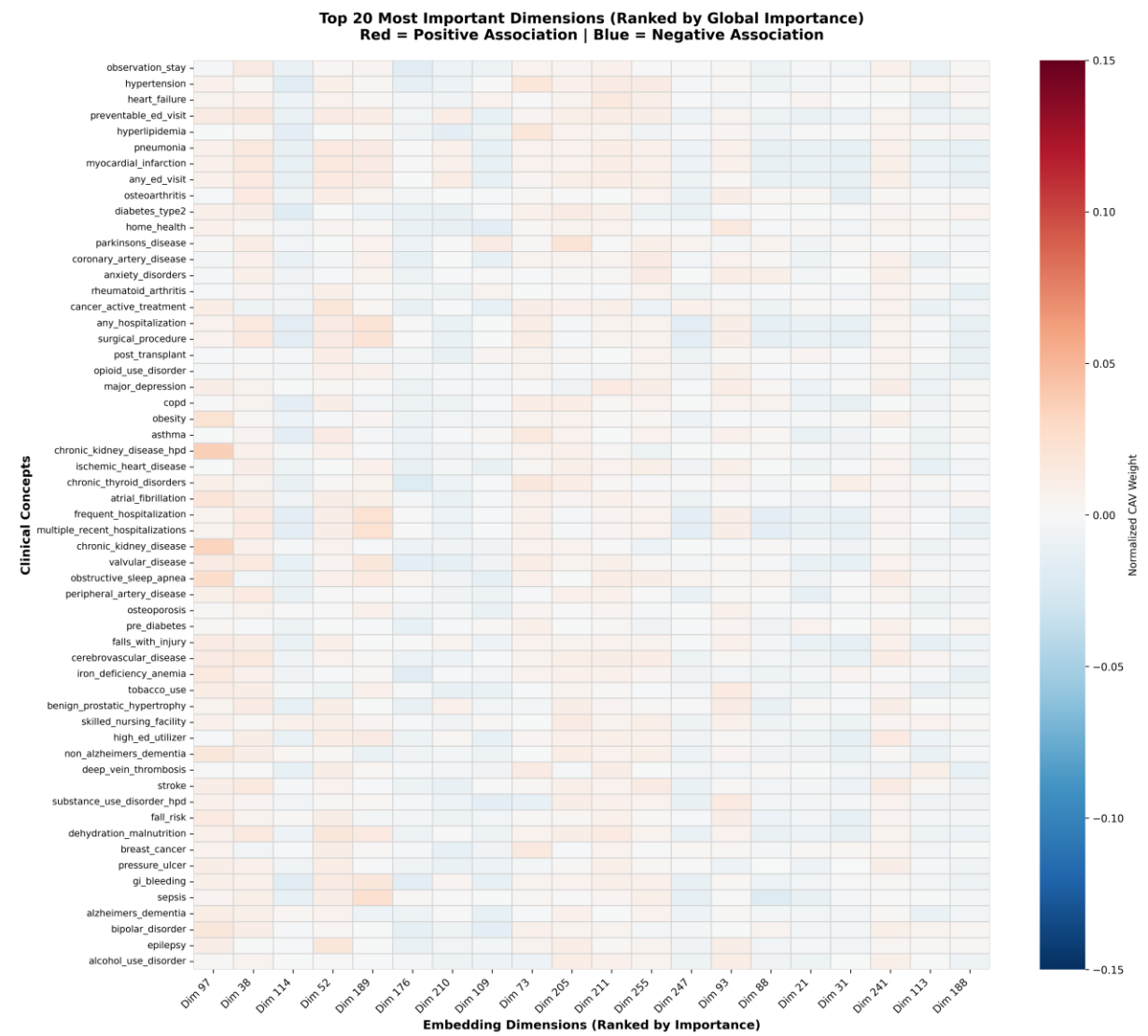
What is CAV?

- Train a linear classifier (logistic regression) to predict each concept from the embeddings
- How to interpret CAV weights:
 - **Positive weight:** Higher dimension value -> Higher probability of concept
 - **Negative weight :** Higher dimension value -> Lower probability of concept
 - **Large absolute weight:** This dimension is important for this concept

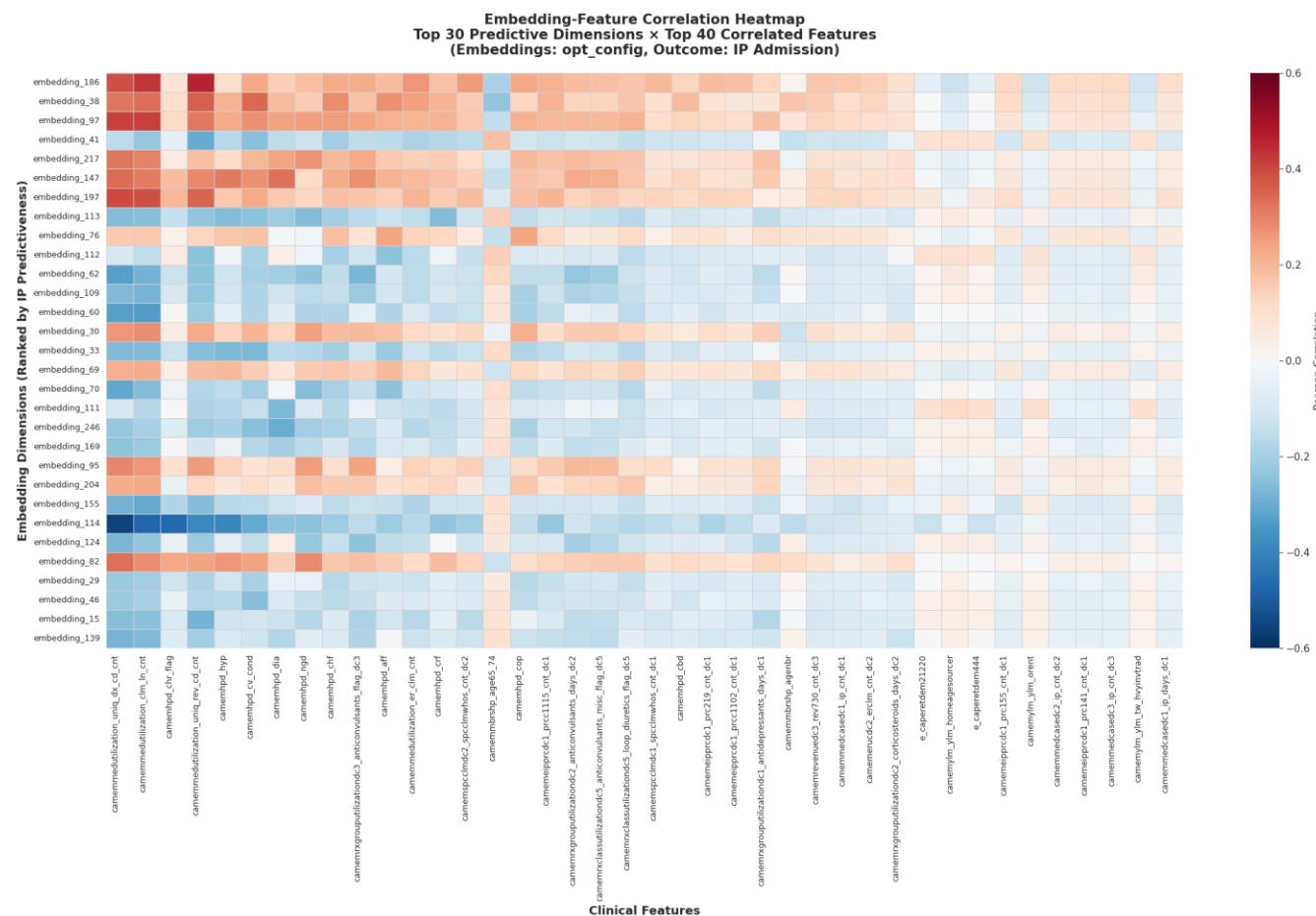
Results (Partial)

- **Dim 97 (Importance 0.492) :** Members with higher values have Chronic Kidney Disease, often with metabolic and respiratory comorbidities (obesity, sleep apnea)
- **Dimension 38 (Importance 0.478):** Members experiencing severe acute medical events such as ER use, IP, surgical procedure, myocardial infarction, pneumonia
- **Dimension 52 (Importance 0.423):** Identifies members undergoing cancer treatment or with severe frailty markers like malnutrition, seizures
- **Dimension 189 (Importance 0.397):** Members experiencing sepsis, frequent IP visits and other surgical procedures

Top 20 Embeddings



Semantic meanings of embeddings with correlation



- **Strong alignment with utilization and disease burden signals**
 - Diagnosis count
 - Total claim lines
 - Revenue code count
 - Chronic condition indicators – Hypertension, Diabetes, CVD
- **Embeddings encode patterns of healthcare interaction**
 - Dimensions reflect overall care intensity and complexity
 - Chronic disease burden (CHF, Hypertension, Diabetes, CV condition) emerge implicitly, which are key risk drivers
- **Embedding_186 (illustrative)**
 - Higher claim volume and diverse service usage
 - Frequent acute utilization (ER visits)
 - Representative of high-touch, more complex members
 - This single dimension captures multi-feature clinical and utilization patterns that would otherwise require many hand-engineered variables

Limitations of Single-Probe Explainability

Previous Approach: Limitations

- Train a single logistic regression probe per clinical concept using embeddings
- Use coefficients to identify important embedding dimensions

Issues Observed

- **No stability validation:** One logistic regression per concept = high variance
- **Arbitrary threshold:** No principled cutoff to determine which dimensions are meaningful
- **Sample sensitivity:** Results changed with different train/test split

New Approach: Bootstrap Concept Activation Vectors (CAVs)

- **Step 1 – Bootstrap Probing**
 - For each concept:
 - Train **100 logistic regression probes**, each trained on a bootstrap sample of the data
 - This produces a **distribution of coefficients** for each embedding dimension rather than a single estimate
- **Step 2 – Statistical Stability Filtering**
 - Criteria:
 - **Sign consistency $\geq 90\%$**
 - **Confidence interval excludes zero**
 - **Coefficient standard deviation ≤ 0.1**
 - **Top 10% by coefficient magnitude**
 - This retains dimensions that show **consistent directional signal** across bootstraps, focusing on **high-impact embedding directions**

High-Quality, Interpretable Embeddings with Actionable Insights

Validation : Stability & Robustness

Quality Metric	Value	Interpretation
Directional Consistency (avg sign-rate)	99.9%	Directions are consistent across bootstrap resamples
Mean Stability (std dev)	0.06	Coefficient estimates show low dispersion across samples
Dimensions with clinical meaning	184/256	72% of dimensions have ≥ 1 stable concept association

Sample Results

Dimension	Clinical Pattern	Example Concepts	Stability
234	Metabolic – Renal	Diabetes (+0.32), Dialysis (-0.31)	100% (2 concepts)
40	SDOH	Financial Strain (+0.13)	99% (1 concept)
211	Cardiovascular	Heart Failure (+0.56)	100% (1 concept)
210	Acute Utilization	ER (+0.91) , Preventable ER (+0.84), Preventative care (-0.32)	100% (7 concepts)

Member-Level Explainability Framework

Method

- **Step 1 – Load member embedding**
 - 256-dimensional vector summarizing clinical history
- **Step 2 – Concept Scoring**
 - For each clinical concept:
 - Apply trained concept detectors
 - Detect whether the embedding contains signals for that concept
 - Produce a concept score
- **Step 3 – Dimension Attribution**
 - Identify which embedding dimensions contributed most to the score
 - Contribution = Member Value x Concept Weight
- **Step 4 – Clinical Interpretation**
 - Link important dimensions to their meanings using the global dimension dictionary

Example: High ER Utilizer

- **Step 1 – Concept Scores**
 - Any ER Visit : 0.92
 - Preventable ER Visit: 0.81
 - Pneumonia: 0.73
 - Poor Care Coordination: 0.58
- **Step 2 – Top Contributing Dimensions**
 - Dimension 210: +0.48, Dimension 52: +0.35, Dimension 29: +0.31
- **Step 3 – Clinical Interpretation (LLM-Enhanced)**
 - **Example output :**
 - *This member demonstrates a high-risk emergency utilization profile, with 92% ED visit likelihood and 81% preventable ED likelihood. The primary driver is Dimension 210, which captures emergency-driven care engagement and inverse associations with preventive behaviors, suggesting reactive rather than proactive healthcare interactions*



Appendix

Dual-Mode Refresh to Optimize Cost, Freshness & Safety

What runs

- **Incremental (Weekly):** Reprocess *only members with recent claims activity*
- **Full (Monthly):** Rebuild *all eligible members* to guarantee completeness

How members are selected

- Uses **last successful run date + 7-day** **lookback buffer**
- Includes members with **medical or pharmacy claims activity**
- **Early filtering** reduces population before heavy processing

Why the buffer matters

- Claims often arrive **days after service date**
- The **7-day buffer** prevents **permanently missing late-arriving claims**

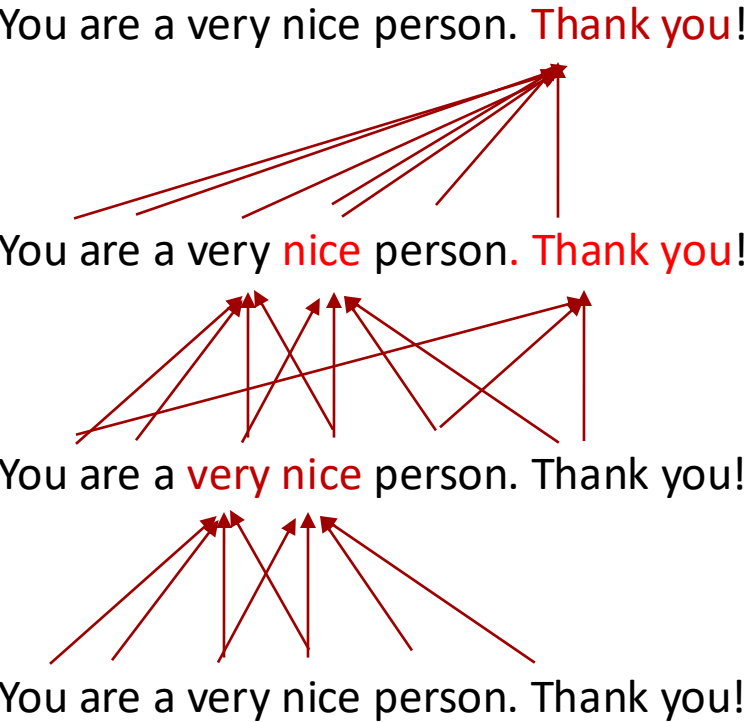
Impact

- **Lower** weekly processing cost
- **Faster** run time
- **Weekly freshness + monthly safety**
- Fully **auditable via control tables**

What is transformer (cont'd)

Transformer = Stacked multi-head attention + position encoding + masking.

Stacked attention

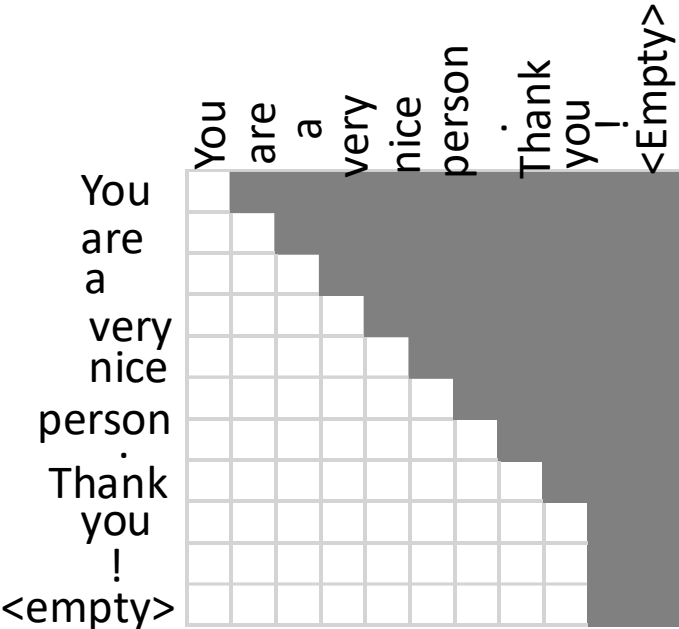


Position encoding

You	0
are	1
a	2
very	3
nice	4
person	5
.	6
Thank	7
you	8
!	9
<empty>	0

Matrix 10xN

Look ahead masking (GPT)



"Attention is all you need" [Vaswani et al. \(2017\)](#)

Our transformer architecture

date	icd9_dx_cd	prcdr_cd	specialty_ctg_nm	plc_srv_ctg_cd	gpi4
2021-01-29	L30.8	99213	Pediatrics	office	6699
2021-01-29	L70.9				6846
2020-10-03	Z23	90460	Pediatrics	outpatient	
		90686			
2020-03-03	J30.9	90460	Pediatrics	outpatient	
	Z00.129	90651			
	Z13.89	90686			
	Z23	96127			
	Z68.52	96160			
	Z71.3	99173			
	Z71.82	99394			
2019-04-28	R10.9	80048	Acute Short Term Hospital	emergency room	
		81001			
		82150			
		83690			
		85025			
2019-04-27	R10.9	74176	Not Mapped	emergency room	
	V86.59XA	74176			
	Y92.838	96365			
	Y93.89	99283			
		99283			
2019-04-19					6131
					6516
2019-04-02	B34.9	87880	Pediatrics	office	
		99214			
2019-01-23	Z00.129	90460	Pediatrics	office	
	Z23	90461			
	Z68.53	90715			
	Z71.3	90734			
	Z71.89	99393			
2018-12-05					120
2018-12-03	J02.9	99203	Missing	outpatient	

Comparing to natural language, our medical data is also in sequences, but more complex:

1. Considering each date is a “word”: there may be multiple data points / types at each date, such as procedure, dx, specialty etc.
2. Dates of the records are NOT continuous.
3. Different medical datasets might not align on dates, such as pharmacy data and claim data .
4. Predicting next ”word” is more challenging than NLP.

Individual_id =49234

Our transformer architecture (cont'd)

Step1. Each date is considered as a word in a sentence. Diagnosis codes, procedure codes etc will be encoded by 1-d CNN.

date	icd9_dx_cd	prcdr_cd	specialty_ctg_nm	plc_srv_ctg_cd	gpi4
2020-03-03	J30.9	90460	Pediatrics	outpatient	
	Z00.129	90651			
	Z13.89	90686			
	Z23	96127			
	Z68.52	96160			
	Z71.3	99173			
	Z71.82	99394			

■
CNN
encoding
↓

■
CNN
encoding
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CNN
encoding
↓

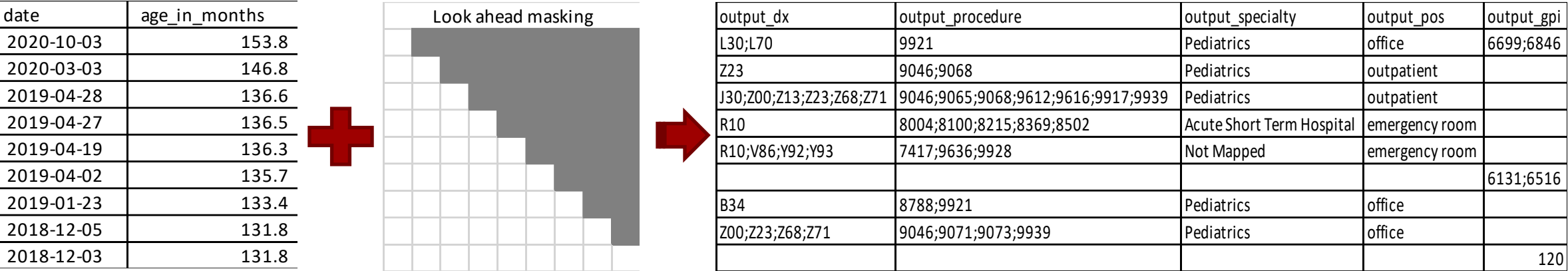
■
CNN
encoding
↓

■
CNN
encoding
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“word” Embedding = A + B + C + D + E

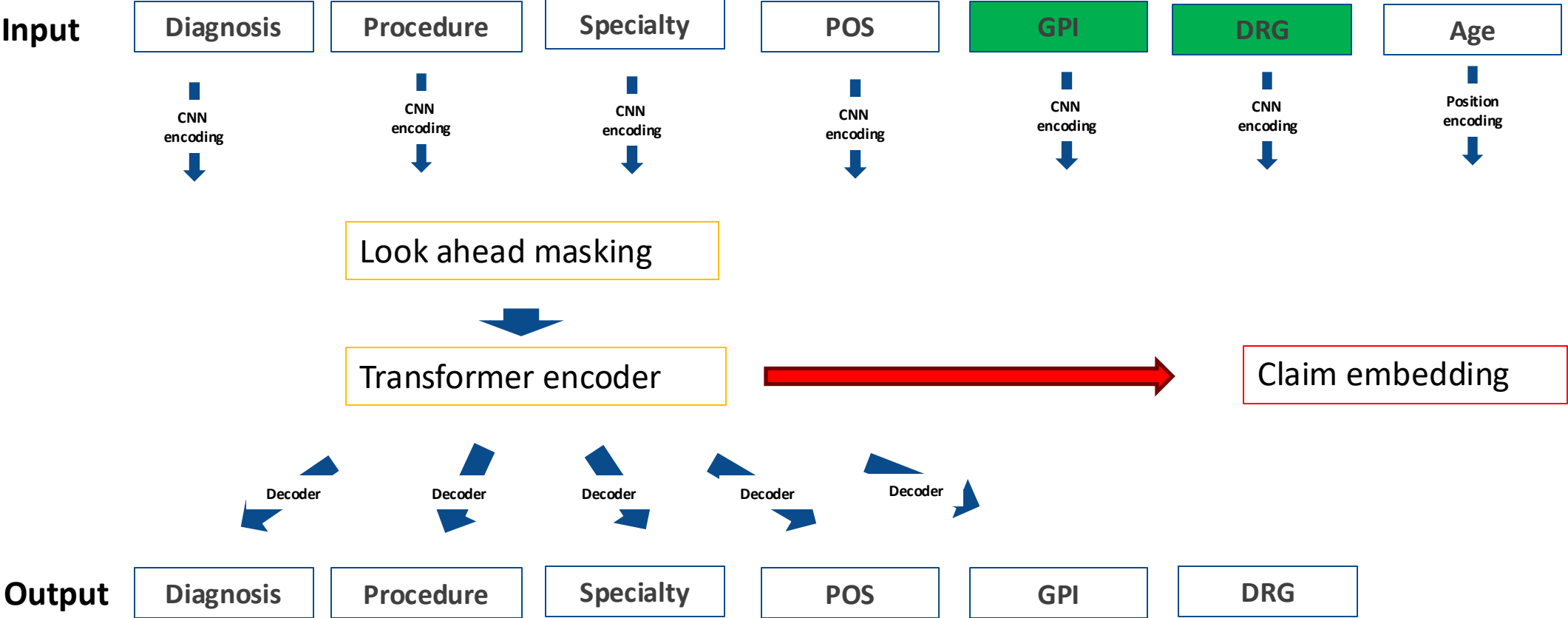
Our transformer architecture (cont'd)

Step2. Use age to encode position. Use look forward masking (like GPT), to predict next medical records, but at higher level. The padded records are ignored in loss calculation.



Our transformer architecture (cont'd)

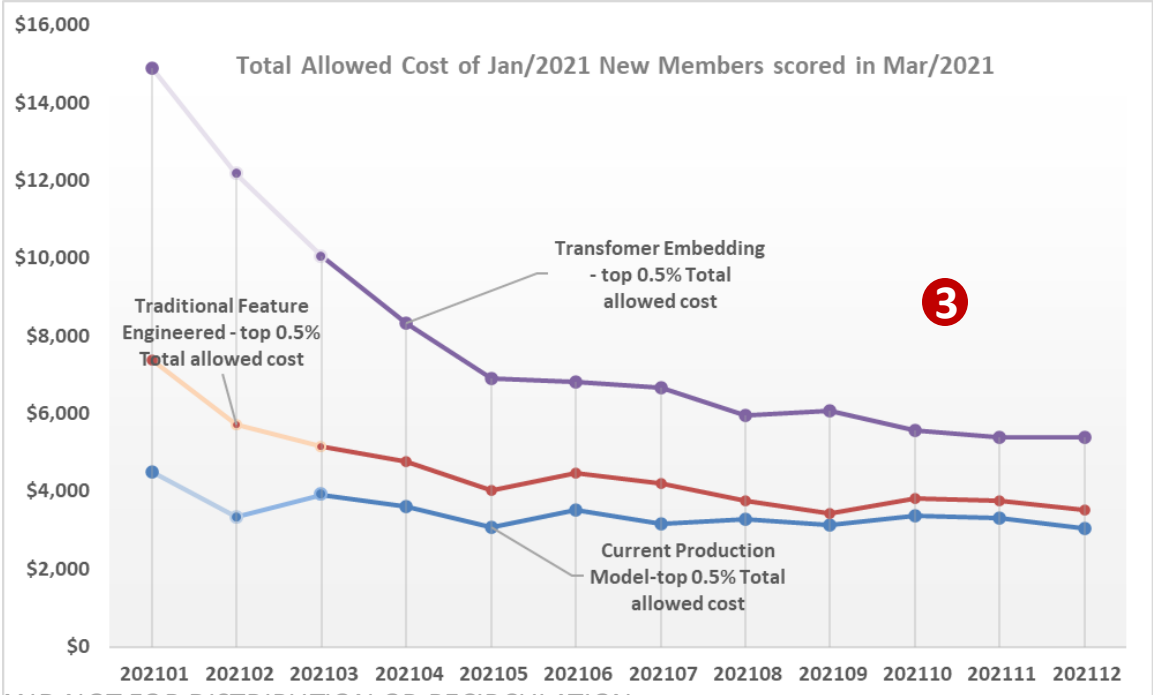
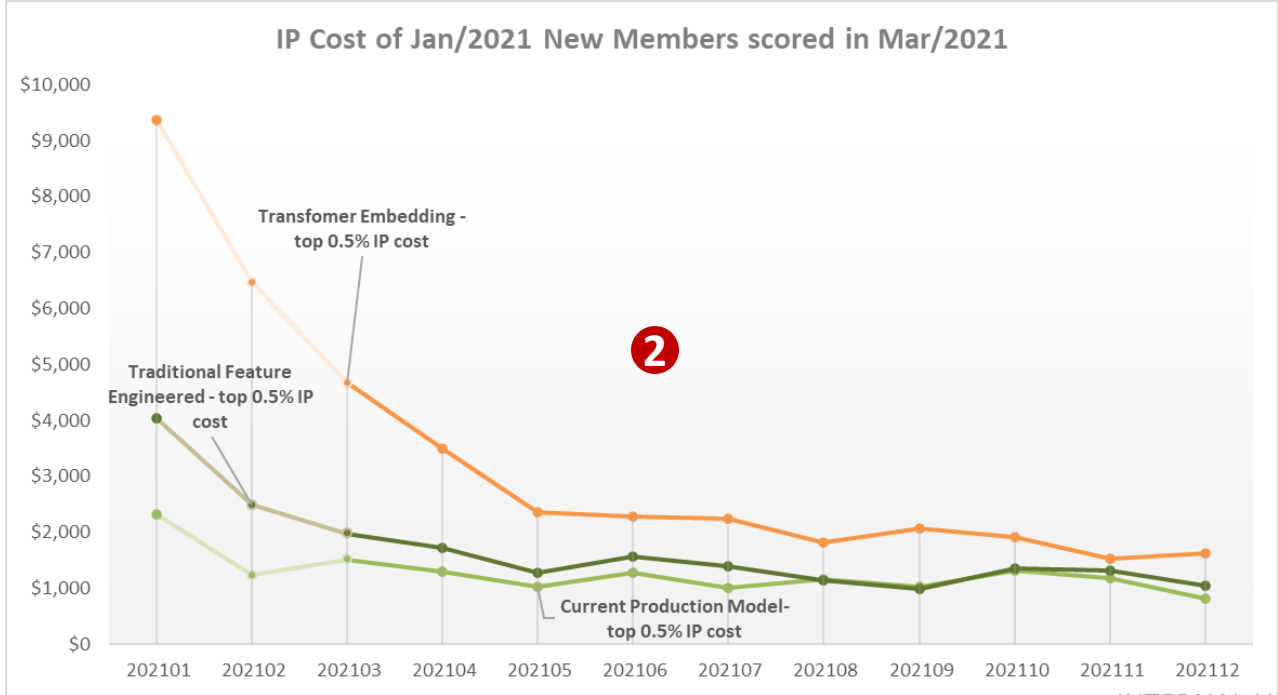
Step3. the layer right before the decoders is the embedding of the medical history, which can be used as predictors in any predictive modeling. So far we have two models: claim embedding and pharmacy embedding.



TE_(v1) model outperforms traditional feature-engineered model by wide margin

Model Lift PPV (% Improvement over Production)			
Member Selection	Benchmark Legacy	Traditional Feature Engineered	1 Transformer Embedding
Top 0.1%	21.4	27.26 (28%)	35.88 (68%)
Top 0.5%	13.5	17.67 (31%)	22.24 (64%)
Top 1%	11.1	13.67 (24%)	16.44 (49%)

- 1
- TE model and traditional feature engineered model perform better than the current model, TE model has the largest lift.
- 2
- TE model identifies members with higher IP likelihood and higher IP/ER and overall medical cost (all cost categories show same pattern)
- 3
- All 3 models indicate that member cost decreases over month after top risk identification, flattening out after month 6, suggesting urgency of care management targeting*
- *please note that due to data delay of the current process, January enrolled new members are scored in March, EDW data up to Feb is used for the March scoring, so we should look at trend starts from March



Medicare IP model with TE (v2)

Comparisons - based on OOT Validation		1%			10%		
	AUC	PPV	Sensitivity	Lift	PPV	Sensitivity	Lift
XGB - NRX V2 (default params)	0.78	49.0%	7.4%	8.16	25.2%	38.3%	4.20
XGB - All Features V2	0.775	48.9%	7.7%	8.15	24.4%	38.6%	4.06
XGB - All Features (default params)	0.775	48.6%	7.7%	8.10	24.3%	38.4%	4.05
Catboost - All Features	0.773	48.5%	7.7%	8.09	24.1%	38.2%	4.02
XGB - RX V2 (default params)	0.771	49.2%	7.9%	8.20	24.0%	38.5%	4.00
XGB - Embeddings + CS (minus Med/RX Claims)	0.771	47.4%	7.5%	7.90	23.9%	37.8%	3.99
XGB - NRX V2 CS Only (default params)	0.771	48.3%	7.3%	8.05	24.3%	36.8%	4.05
XGB - CS Features V2	0.768	47.3%	7.5%	7.88	23.5%	37.3%	3.92
Catboost - Embeddings + CS (minus Med/RX Claims)	0.768	47.1%	7.5%	7.86	23.8%	37.7%	3.97
XGB - CS Only (default params)	0.766	47.0%	7.4%	7.83	23.4%	37.0%	3.90
Catboost - CS Only	0.765	46.7%	7.4%	7.79	23.4%	37.0%	3.90
XGB - RX V2 CS Only (default params)	0.764	46.9%	7.5%	7.82	23.0%	34.8%	3.83
XGB - Embeddings (v2) Only	0.748	45.1%	7.1%	7.39	23.4%	36.6%	3.82
XGB - HPD Only (default params)	0.738	-	-	-	-	-	-
Production	0.732	38.1%	6.1%	6.10	20.9%	33.2%	3.34
XGB - Embeddings (v1) Only	0.717	35.8%	5.6%	6.20	20.4%	31.9%	3.36

- Embeddings did not perform as well as traditional feature engineering here, due to one key problem- RX (GPI4) was absent from the embeddings, which is crucial for the Medicare population.
- We would expect a boost to performance for embedding-only models, and perhaps a small boost to combined feature models.

Medicaid IP model with TE_(v2) preliminary findings

- The TE model was trained in commercial and Medicare but still performs well in Medicaid
- Current best model in development has an AUC of ~0.87 and lift of ~20 in top 1% (data not shown)

25% holdout embeddings vs. Product Spec (Medicaid)		1%			10%		
	AUC	Precision	Recall	Lift	Precision	Recall	Lift
Logistic Regression - Embeddings only (6 month)	.84	28.9%	17%	16.98	10%	58.1%	5.81
Production (12 month)		45.1%	10.3%	10.29	19.6%	44.6%	4.46

- Breakout by tenure suggests a new member model in Medicaid is not necessary

Logistic Regression – Embeddings only (by membership length)		1%			10%		
	N (%)	Precision	Recall	Lift	Precision	Recall	Lift
0-3 Months	35,354 (6.2%)	19.8%	10.8%	10.79	6.8%	36.9%	3.69
4-8 Months	55,853 (9.8%)	26.7%	17.8%	17.77	8%	53.5%	5.35
9+ Months	477,535 (84%)	29.9%	17.8%	17.83	10.3%	61.6%	6.16

TE empowers data scientists to build better models more efficiently

Better performing model

- **Better lift/precision/AUC**
- **TE captures the intricacy of our data**, both structured and unstructured in a sequential manner, generating a condensed representation that encompasses causal effects and enhances our predictive models. This will be harder to achieve manually with thousands of sparse features. Traditional feature selection often drops features with low prevalence, such as pancreatic cancer, TE can capture rare and costly diseases.
- **Bigger business impact** Early experiments showed TE model identifying higher risk members (more IP/ER utilization, higher spend)

More Efficient

- **Time-saving**. As a data scientist, we spend up to 80% of our time cleaning and transforming data to generate actionable insights and build machine learning models.
- **Standardizes and streamlines process with claim history data**. It simplifies data scientists' work, saves **weeks**' time and system resources, allowing us to focus on applying the latest ML skills and building models that effectively address our business goals, solve problems, and deliver value to our customers.

Retraining Model – Architecture Redesign and Resource (Hypothetical cost computation)

Nvidia GPU configuration (Effective performance, TeraFLOPs/sec) ^{2,3} - hourly rate ¹	Original Model Total params: 27.7M Total FLOPs for training: 2.142 petaFLOPs ⁴		Proposed Model Total params: 34.8M Total FLOPs for training: 2.147 petaFLOPs	
	Total estimated time	Total cost	Total estimated time	Total cost
4* T4 (41.6) - \$2.99/h	1180.8 h	\$3520	902.4 h	\$1728
4* L4 (111.3) - \$3.3/h	355.2 h	\$1184	217.6 h	\$704
4* A100 (561.6) - \$25/h	89.6 h	\$2208	41.6 h	\$1072
4* H100 (1820) - \$36.38/h	24.32 h	\$896	12.8 h	\$480

Training setup: 16M members, 2 epochs
Experimentation setup: 1-5M members, 1-2 epochs

¹ The GPU price rates are estimated by vertex AI workbench instance.
² FLOPs = Floating-point operations
³ One teraFLOPs = 1 * 10¹² FLOPs
⁴ One petaFLOPs = 1 * 10¹⁵ FLOPs

Retraining Model – Architecture Redesign and Resource (Adjusted based on real ablation running on T4)

Nvidia GPU configuration (Effective performance, TeraFLOPs/sec) ^{2,3} - hourly rate ¹	Original Model Total params: 27.7M Total FLOPs for training: 2.142 petaFLOPs ⁴		Proposed Model Total params: 25.33M Total FLOPs for training: 2.147 petaFLOPs		GCP official estimates
	Total estimated time	Total cost	Total estimated time	Total cost	
4* T4 (65) - \$2.99/h	255.5 h (10.6 d) + 72 h	\$979	108.3 h (4.5 d) + 72 h	\$539	\$1275
4* L4 (242) - \$4/h	141.9 h (5.9 d) + 72 h	\$856	60.2 h (2.5 d) + 72 h	\$529	\$561
8* L4 (242) - \$8/h	79.8 h (3.3 d) + 72 h	\$1,214	33.8 h (1.4 d) + 72 h	\$846	\$961
4* H100 (1,979) - \$25/h	34 h (1.4d) + 72 h	\$ 3,938	8.6 h (0.4 d) + 72 h	\$3,376	\$3,540
8* H100 (1,979) - \$88.49/h	17 h (0.7 d) + 72 h	\$7,876	4.3 h (0.2 d) + 72 h	\$6,752	\$6,723

* The results are estimated from experimentation setup on T4 (**1.7M members, 1 epochs**); the formal training is going to involve 16M members with at most 4 epochs; the projected training time and cost for L4 and H100 are derived from scaling performance against T4.

¹ The rates are on-demand GPU price estimated by vertex AI workbench instance.

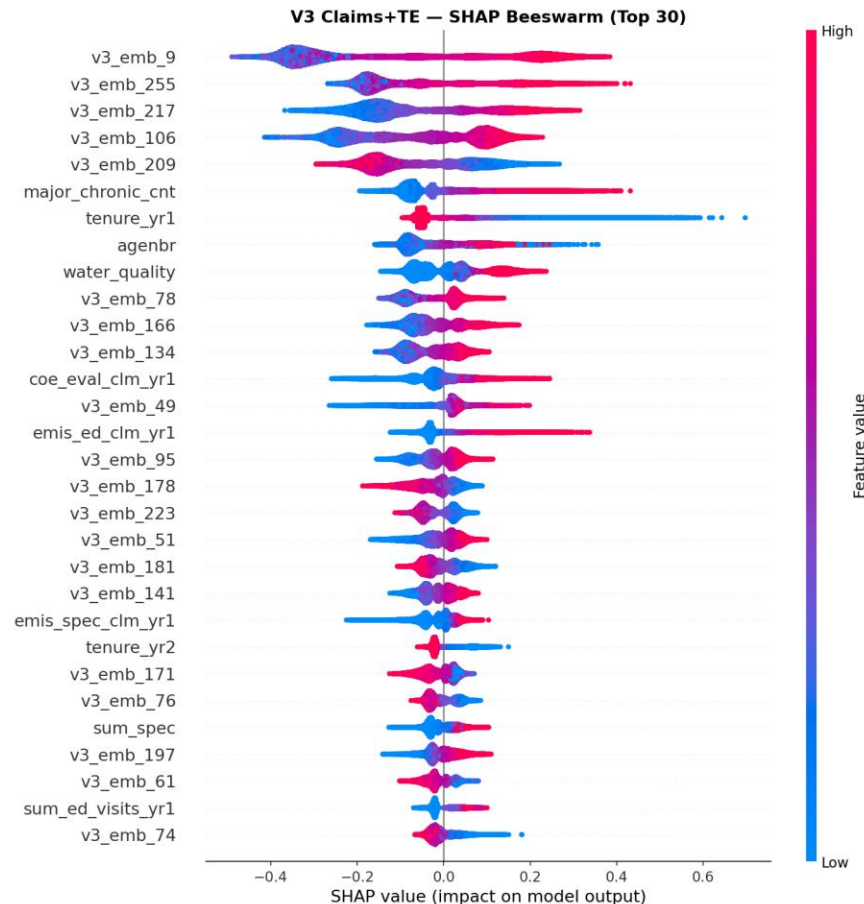
² FLOPs = Floating-point operations

³ One teraFLOPs = 1 * 10¹² FLOPs

⁴ One petaFLOPs = 1 * 10¹⁵ FLOPs

Top embedding drivers in Medicaid IP align with meaningful clinical risk patterns

Embedding-derived features dominate the top SHAP drivers in the hybrid Claims + TE model, and early interpretation suggests these dimensions encode clinically meaningful signal rather than noise



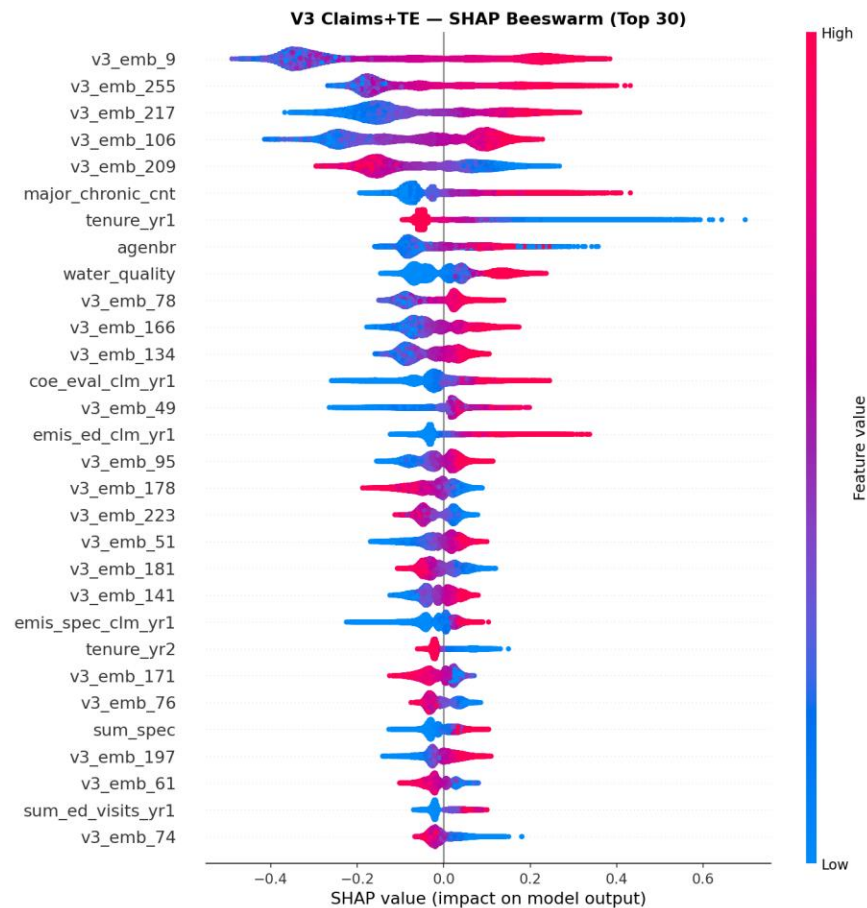
Example: v3_emb_9

Appears to encode a **chronic adult profile with sustained pharmacotherapy and elevated hospitalization risk**

- Major chronic count: **4.3 vs 0.2**
- Unique drugs: **8.4 vs 0.7**
- Maintenance fills: **31.1 vs 0.1**
- IP admission rate: **6.3% vs 0.3%**
- Mean age: **63 vs 8**

Top SHAP drivers are dominated by embedding features

Embedding-derived features dominate the top SHAP drivers in the hybrid Claims + TE mode



Downstream evaluation (small scale) – Follow up ablations (cont.)

Major findings:

1. The loss-based and frequency-based solutions for relieving gradient starvation did not generate significant improvements in downstream evaluations.
2. Increasing the embedding dimension to 512 and training size improve the lift@1% by **6.85%** and **14.44%**

	Production IP model	Optimized-FA-TE		Optimized-FA-TE-Focal loss		Optimized-FA-TE-Focal loss + Density sampling batch		Optimized-FA-TE-512dim		Optimized-FA-TE- M	
TE Dimension	NA	256		256		256		512		256	
TE Training size	NA	1.75M		1.75M		1.75M		1.75M		5.7M	
Feature type	Engineered feature	Hybrid	Embedding	Hybrid	Embedding	Hybrid	Embedding	Hybrid	Embedding	Hybrid	Embedding
Medicare IP model											
AUC ROC	0.742	0.758	0.736	0.753	0.736	0.751	0.738			0.756	0.750
AUC PR	0.199	0.21	0.186	0.202	0.191	0.198	0.191			0.212	0.197
Lift@1%	6.201	6.391	5.742	6.19	6.06	6.182	5.683			6.689	5.981
Commercial IP model											
AUC ROC	0.832	0.824	0.79	0.821	0.791	0.815	0.783	0.824	0.798	0.833	0.8
AUC PR	0.1	0.105	0.078	0.091	0.069	0.088	0.069	0.092	0.071	0.114	0.08
Lift@1%	18.682	19.082	15.059	18.264	14.746	17.612	14.646	18.37	16.09	19.44	17.233

Product vision

Our goal: build out explainability tools for both Transformer embedding (TE) and leverage LLMs to reduce the effort needed to effectively communicate our explainability findings.

